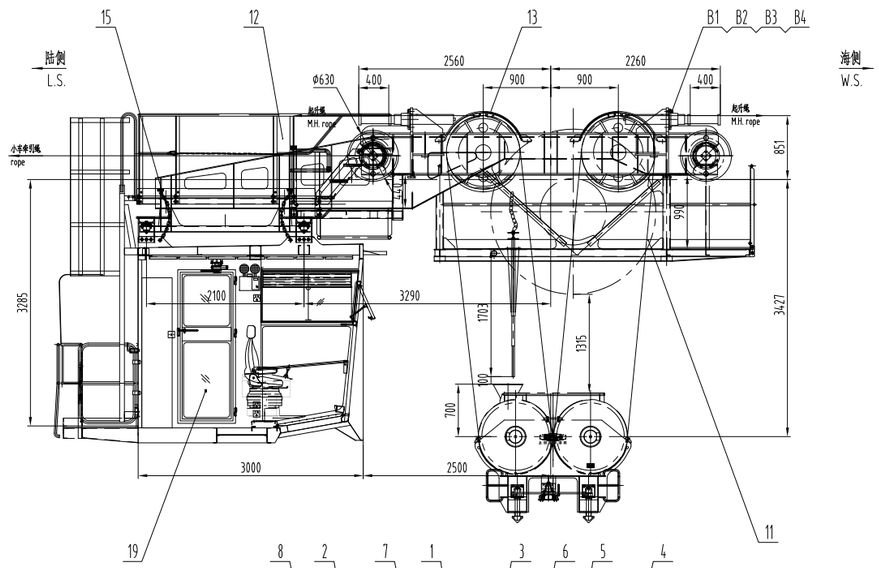
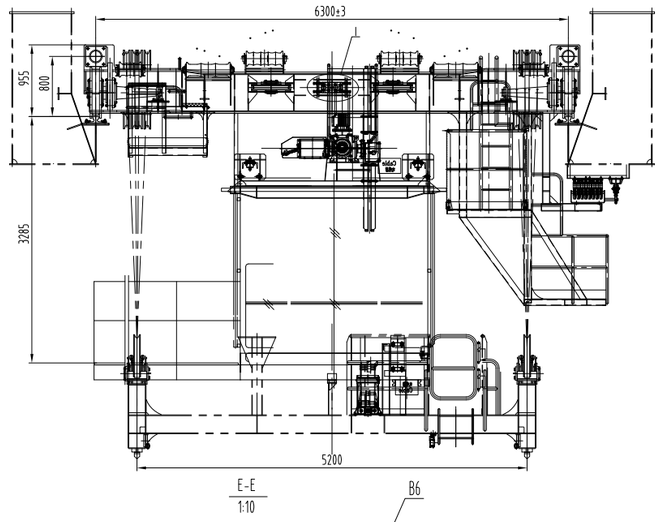
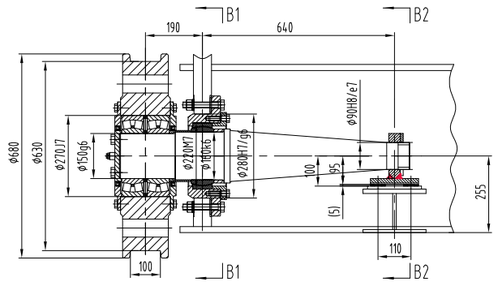
rev	date	by	description
01	2012.05.30	hxl	
02	2012.08.17	hxl	





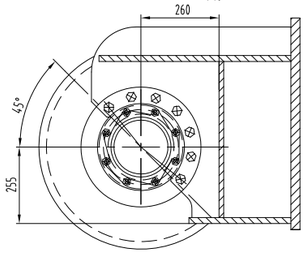
E-E
1:10

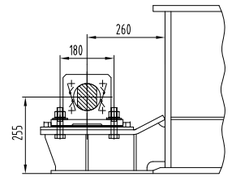


B-B
1:10

B1-B1
1:10

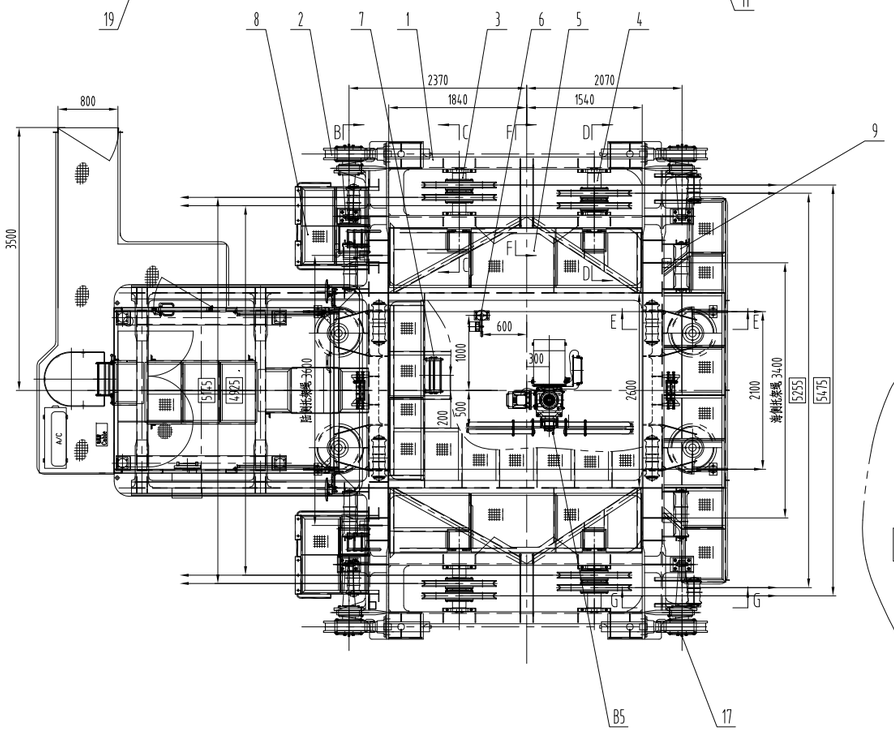
B2-B2
1:10

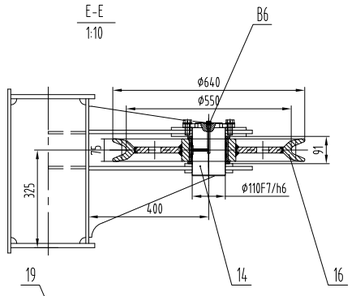


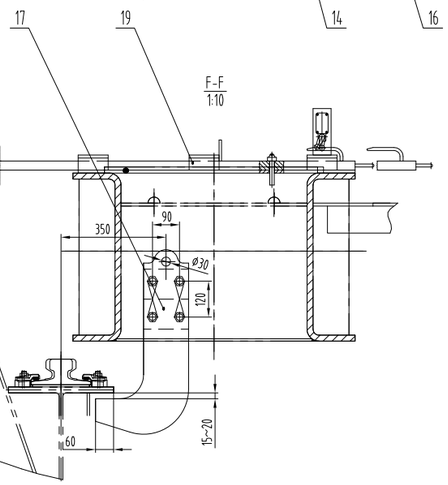


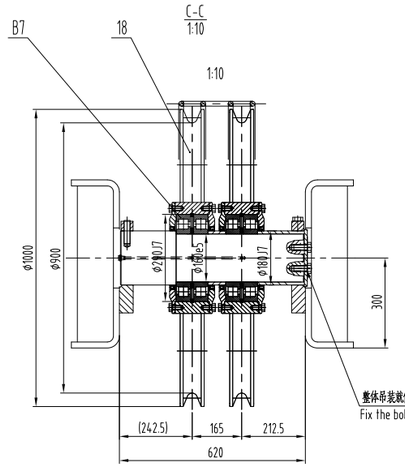
C-C
1:10

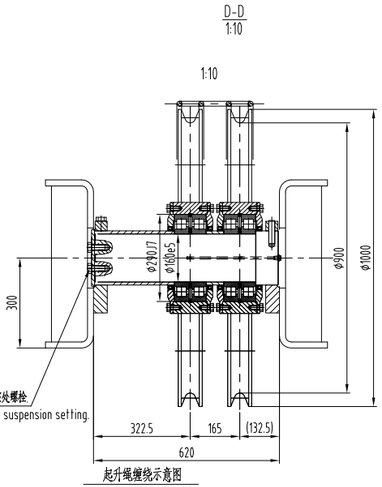
D-D
1:10



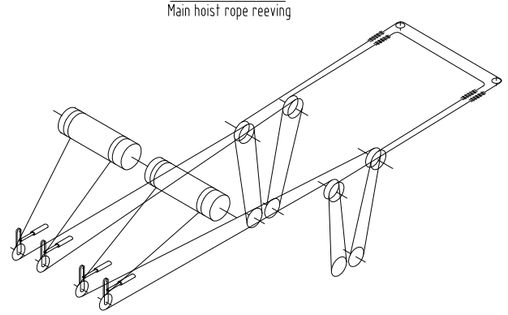




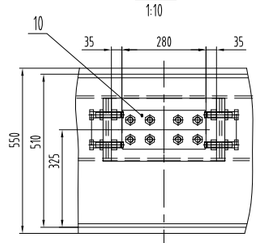




起升绳缠绕示意图
Main hoist rope reeving



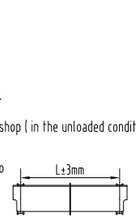
Detail I
1:10



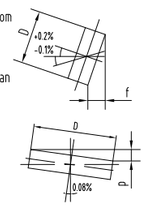
centerline of trolley
小车中心线 ℄

Technical Requirements 技术要求

1. Tolerance of Trolley wheels installaion in workshop (in the unloaded condition):
在装配车间车轮排装的偏差要求:
   a. The tolerance of span L of trolley related to wheel centre is not more than ±3mm.
   小车车轮中心间的跨距 L 偏差≤±3mm.
   b. The axle bores of wheels opposite to each other at each side of the track, the height tolerance in the vertical plane less than 2mm of the wheel centre distance;
   两侧轨道上相对车轮轴孔轴线在垂直面内的高度方向的偏差≤2mm;
   c. The centre planes of wheels rolling on a common rail must not diverge more than 2mm from the theoretical axis;
   同一轨道上的车轮中心面同车轮理论中心线的偏差≤2mm;



   d. In cases where the top of the rail is flat,the inclination of the wheel axis from the horizontal, for the unladed crane,must be between +0.2% and -0.1% ;
   车轮轴线相对于水平面的倾斜度,必须在+0.2%~-0.1%之间;
   e. The axes of the wheel bores must not have an angular deviation greater than ±0.08% from its theoretical axis,in the horizontal plane.
   轮孔中心线同其理论中心线之间在水平面内的角度偏差,不得大于±0.08%.



车轮直径(D) 检测值	Φ500	Φ630	Φ710
f (mm)	+1.0~-0.5	+1.26~-0.63	+1.42~-0.71
D (mm)	±0.4	±0.504	±0.568

2. The fastening force and moment of every nut follow the parts drawing .
各零部件中紧固件的予紧力及予紧力矩详见各零部件装配图.
3. The fastening force of M24 is 151.9 kN, and the pretightening torque is 530 Nm(k=0.145) .
M24 螺栓的预紧力为151.9kN, 预紧力矩为530Nm(k=0.145) .

PROJECT: SliLanka CICT 12 STS Cranes				DESIGN STAGE:			SCALE: 1:40	
ITEM NAME: 小车总成Trolley Arrangemant				DRAW NO: J526A0200			PROJ.	
ZPMC	DSGN			TRACE			APP.	
	DRAW			CHCK			SET/CRANE	
	CHCK			VER.			WEIGHT	

明 细 表
SUBSIDIARY LIST

PAGE: 第 / 页

*The quantity and weight of this list is for one set

*此表为一套的数量和重量

设计：

22438 kg/set, 1 set/crane, 22438 kg/crane

校对：

PROJECT NAME or CODE: Sri Lanka CICT 12 STS Cranes			SUB-PART NAME: 小车总成			SUB DRAW NO: J526A0200	
序号 NO.	图号或标准 SUB-NO.or STDD	名称 NAME	规格或型号	材料 MTRL	每套数量 Q'TY/SET	单重 PIECE WGHT	总重 TOTAL WGHT
1	J526A0201	Trolley structure 小车架		Welded 焊接件	1	7880	7880
2	ZT2-020936-00	Trolley Wheel Ass. 小车车轮装配		Ass. 组件	4	550.1	2200.4
3	ZT2-021023-00	Main hoist Sheave shaft Ass. 1 起升滑轮轴装配1		Ass. 组件	2	135.7	271.4
4	ZT2-021123-00	Main hoist Sheave shaft Ass. 2 起升滑轮轴装配2		Ass. 组件	2	135.8	271.6
5	J526A0205	Walkway & Platform 走道平台		Ass. 组件	1	670	670
6	J526A0206	M.H. Limit Switch 重锤限位		Ass. 组件	1	20.6	20.6
7	J526A0207	Ladder 直梯		Ass. 组件	1	34.5	34.5
8	J526A0208	L.S. Platform 陆侧维修平台		Ass. 组件	1	518	518
9	J526A0209	Idler Ass. 托辊装配		Ass. 组件	1	334.8	334.8
10	ZT2-021615-00	Clamp Plate Ass. 压绳板装置		Ass. 组件	2	51.3	102.6
11	J526A0211	Spreader Cable Reel Arrangement 吊具电缆卷盘支撑架		Ass. 组件	1	1245	1245
12	J526A0212	Ladder & Platform on the Cab Roof 司机室顶部斜梯&平台		Ass. 组件	1	227	227
13	ZT2-021603-J526A	Hoist Sheave Cover &Closed Plate Φ900起升滑轮罩壳及封板		Ass. 组件	1	206.8	206.8
14	ZT2-021307-00	Equalizer Sheave shaft Ass. 牵引滑轮轴装配		Ass. 组件	4	10.6	42.4
15	J526A0215	Chain & Discharge 链条卸扣		Ass. 组件	4	3	12
16	ZT2-012932-00	Equalizer Sheave均衡滑轮		Ass. 组件	4	54	216
17	ZT2-021634-00	Mousing Hook 防跳装置		Ass. 组件	2	32.8	65.6
18	ZT2-013069	Sheave ass.滑轮装配		Ass. 组件	8	184.1	1472.8
19	J526A0213	trolley anchor小车锚定		Ass. 组件	1	65	65
20	J526A3100	Trolley Operator's Cab 司机室		Ass. 组件	1	3342	3342
21		Electric 电气		Ass. 组件	1	2000	2000

明 细 表
SUBSIDIARY LIST

PAGE: 第 / 页

*The quantity and weight of this list is for one set

*此表为一套的数量和重量

设计：

22438 kg/set, 1 set/crane, 22438 kg/crane

校对：

PROJECT NAME or CODE: Sri Lanka CICT 12 STS Cranes			SUB-PART NAME: 小车总成			SUB DRAW NO: J526A0200	
序号 NO.	图号或标准 SUB-NO.or STDD	名称 NAME	规格或型号	材料 MTRL	每套数量 Q'TY/SET	单重 PIECE WGHT	总重 TOTAL WGHT
B1	5/400 BQD140-407	Buffer 缓冲器		Ass. 组件	4	76	304
B2	GB32.1-88	Bolt 螺栓	M24x70	8.8s 达克罗	16		
B3		Stainless Wire 不锈钢丝	⌀2		若干		
B4	GB97.1-85	washer垫圈	24	200HV	16		
B5		Spreader cable reel 吊具电缆卷盘		Ass. 组件	1	500	500
B6	JISB 0203	油嘴 PT1/4		黄铜	20	0.2	4
B7	SKF	轴承 NJ232EC	⌀290x⌀160x48		16	14	224

View A

View B

Wheel 车轮中心线

C-C
1:10

Piece 19
1:10

c1-c1
1:10

View P

Piece 13
1:10

Piece 9
1:10

Piece 95
1:8

F-F
1:10

Piece 15 对称制作
1:10 Symmetry

D-D
1:20 (Rotate 90°)

p1-p1
1:10

View p2
1:10

f1-f1
1:10

G-G
1:10

Piece 42
1:10

View H
1:10

Piece 20
1:20

d2-d2
1:20

Piece 26
1:10

Piece 90
1:10

J-J
1:10

Piece 31
1:10

Piece 36
1:10

Piece 91
1:10

Detail I
1:10

Piece 28
1:20

Piece 7
1:20 对称制作
Symmetry

Piece 1
1:20 对称制作
Symmetry

i1-i1
1:10

View j2
1:10

Detail III
1:3

Detail IV
1:2

Piece 34
1:1

Piece 41
1:10

Piece 83
1:10

Piece 92
1:10

The Comparison Table of Welding Symbols	
AWS	ZPMC

Minimum Fillet Weld Size 角焊缝最小尺寸	
Base Metal Thickness 基本金属厚度 (mm)	Minimum Size of Fillet Weld(mm) 角焊缝最小尺寸
T≤6.4	3
6.4<T≤12.7	5
12.7<T≤19.0	6
19.0<T	8

	J526A0201(1/3)	Trolley frame 小车架	Welded 焊接件				1:40
ZPMC	DRAW		TECH.			TRACE	
	CHCK		STDD			CHCK	

M-M 1:20

spreader centerline 吊具中心线

n1-n1 1:20

n2-n2 1:20

n3-n3 1:20

View I1 1:10

View I2 1:10

L-L 1:10

M2-M2 1:20

M1-M1 1:10

Piece 54 1:10

View M6

Piece 47 1:10

View I3 1:10

Detail UI 1:5

Detail UII 1:5

Detail UIII 1:5

U1-U1 1:10 Rope clamp stiffener 压绳板装置筋板

L2-L2 1:10

L1-L1 1:10

Piece 49 1:10

Piece 55 1:10

M3-M3 1:10

M7-M7 1:10

Piece 65 1:20

Piece 62 1:10 FCM

M4-M4 1:20

M5-M5 1:20

Detail V 1:5

Piece78 1:10

U-U 陆侧梁 1:20 Land side

U3-U3 1:10

Detail H 1:10

Piece 5 1:20

Piece 6 1:20

Piece 59 对称制作 symmetry 1:20

Piece 85 1:10

Piece 14 1:20

Piece 81 1:10

u4'-u4' 1:10

U4-U4 托架小车轨系固筋板 1:10 catenary trolley rope tie plate stiffener

The Comparison Table of Welding Symbols

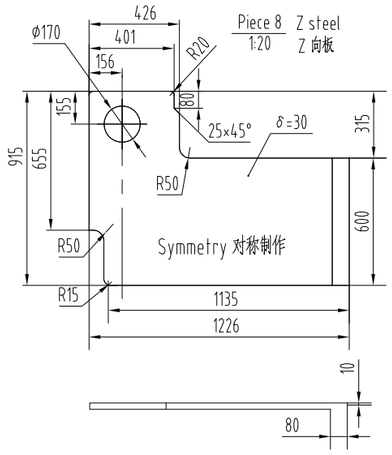
AWS	ZPMC

Minimum Fillet Weld Size 角焊缝最小尺寸

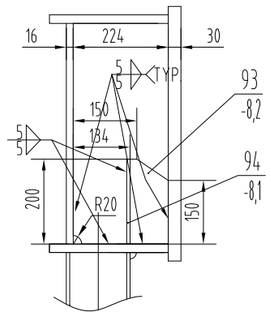
Base Metal Thickness 母材厚度(mm)	Minimum Size of Fillet Weld(mm) 角焊缝最小尺寸
T≤6.4	3
6.4<T≤12.7	5
12.7<T≤19.0	6
19.0<T	8

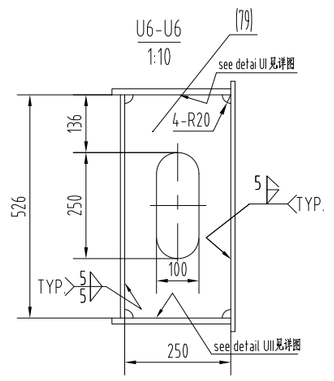
	J526A0201(2/3)	Trolley frame 小车架	Welded 焊接件		8060.68	1:40			
ZPMC	DRAW			TECH.			TRACE		
	CHCK			STDD			CHCK		

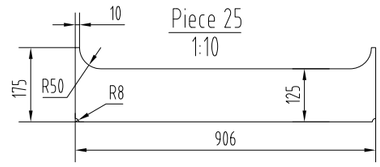
rev	date	by	description
01	2012.05.30	hxl	
02	2012.08.17	hxl	



托令牵引筋板 U5-U5
Festoon plate 1:10







技术要求

1. 焊条为E5015.
2. 对接焊缝必须100%UT检查.
3. 其它未注焊缝参考同类焊缝,焊高用AWS要求的最小焊缝高度.
4. 4小车架两侧密封的箱体做气密性试验,试验压力为1.5pi(0.01Mpa).
5. 明细表尺寸供参考,实际以放样为准,放样图下料留因校核尺寸.

Technical requriment

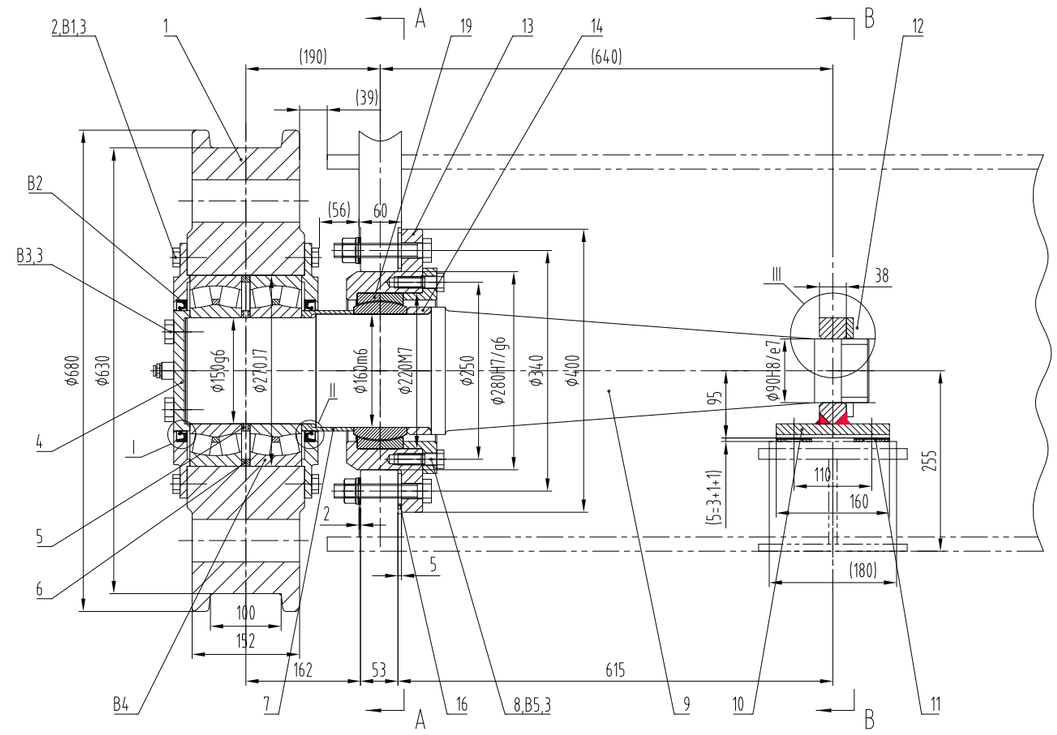
1. Welding electrode is E5015.
2. Butt weld must be inspected in 100.
3. The quality standard of weld should attain to AWS. The other weld line (no drawing) may be ref. the same type,Where fillet weld size is not show,use minimum fillet weld per AWS.
4. Sealing test should be done at two-side boxes of trolley frame.The ssure is 1.5psi(0.01MPa).
5. Demensions in the list are for reference only.

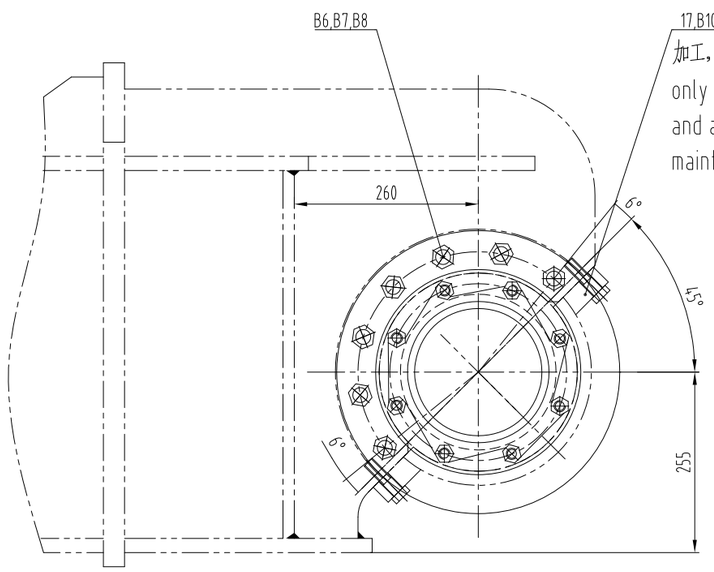
NO.	SUB-NO. or STDD	NAME & SIZE	MTRL or STYLE	Q'TY/SET	SET*WEIGHT		SCALE
98		焊缝重量 solder weight		1		163	
97		-8×80×836	Q345B	4	3.6	14.4	
96		-8×80×588	Q345B	4	2.6	10.4	
95		-3×80×106	Q345B	4	0.15	0.6	
94		-8×84×200	Q345B	1	0.1	0.1	
93		-8×200×224	Q345B	2	0.5	1	
92		-20x160x350	Q345B	4	3.9	15.6	
91		-20x336x630	Q345B	4	14.8	59.2	
90		-10×340×310	Q345B	4	6.2	24.8	
89		-10x80x120	Q345B	16	0.7	11.2	
85		-10x340x1360	Q345B	2	25.8	51.6	
84		-3×30×1368	Q345B	2	0.9	1.8	
83		-14×149×224	Q345B	8	2.9	23.2	
82		-8x175x500	Q345B	2	3.7	7.4	
81		-20x125x270	Q345B	4	4.0	16	
80		-10x50x115	Q345B	8	0.5	4	
79		-12x250x526	Q345B	4	9.6	38.4	
78		-8x250x486	Q345B	8	5.3	42.4	
77		-8x80x280	Q345B	8	1.2	9.6	
76		-8x80x490	Q345B	4	2.2	8.8	
75		-10x400x3595	Q345B	2	56	112	
74		-6x100x1632	Q235	2	8.1	16.2	
73		-8x66x196	Q345B	12	0.8	9.6	
72		-6x150x150	Q235	4	0.5	2	
71	24.5kg/m	方钢管 140x140x5-1580	Q235	2	40.1	80.2	
70		-8x200x2266	Q345B	2	28.5	57	
69		C20x75x9-2266	Q235	2	58.4	116.8	
68		-8x130x2400	Q345B	1	19.6	19.6	
67		-8x440x2910	Q345B	1	44	44	
66	FCM	-8x184x228	Q345B	8	2.6	20.8	
65		-10x280x636	Q345B	4	7.0	28	

NO.	SUB-NO. or STDD	NAME & SIZE	MTRL or STYLE	Q'TY/SET	SET*WEIGHT		SCALE
64		-8x120x130	Q345B	8	1.0	8	
63		-8x120x506	Q345B	8	2.7	21.6	
62	FCM	-8x228x320	Q345B	8	3.3	26.4	
61		-8x80x729	Q345B	2	3.2	6.4	
60		-8x80x703	Q345B	2	3.1	6.2	
59	Symmetry 对称制作	-10x740x3424	Q345B	2	64.7	129.4	
58		-8x80x431	Q345B	2	2.2	4.4	
57		-8x80x403	Q345B	2	2.0	4	
56		-8x942x3642	Q345B	2	72.4	144.8	
55		-8x80x200	Q345B	8	0.9	7.2	
54		-12x120x599	Q345B	2	3.8	7.6	
53		-8x80x573	Q345B	2	2.4	4.8	
52	FCM	-10x200x3092	Q345B	2	48.2	96.4	
51		-8x940x3160	Q345B	2	92.3	184.6	
50		-8x80x301	Q345B	2	1.5	3	
49		-12x120x273	Q345B	2	2.2	4.4	
48		-20xΦ90	Q345B	4	1.0	4	
47		-12×310×240	Q345B	4	4	16	
46		-10×308×264	Q345B	4	6.4	25.6	
45		-12×308×264	Q345B	4	7.7	30.8	
44		-16×336×415	Q345B	4	17.5	70	
43		-20x80x100	Q345B	4	1.5	6	
42		-12x170x390	Q345B	4	4.3	19.6	
41		-20x196x465	Q345B	4	12.3	49.2	
40		-10x400x264	Q345B	4	8.3	33.2	
39		-14×400×315	Q345B	8	8.8	70.4	
38		-10×270×290	Q345B	4	4.2	16.8	
37		Φ12-109	Q235B	4	0.1	0.4	
36		-6x167×266	Q235	8	1	8	
35		-6x109×563	Q235	4	2.4	9.6	
34		-6x48x50	Q235	16	0.1	1.6	
33		-12x545x590	Q345B	8	20.5	164	
32		-(9)xΦ160xΦ110	Q345B	8	0.9	7.2	
31		-10x436x470	Q345B	8	8.3	66.4	
30	J526A020101	Rope Support 托绳架	Ass. 组件	4	4.2	16.8	
29		-(9)xΦ260/Φ110	Q345B	8	2.8	22.4	
28		-8x600x175	Q345B	8	4.8	38.4	
27		-10x250x432	Q345B	8	7.9	63.2	
26		-16x310x423	Q345B	4	11	44	
25		-8x175x906	Q345B	4	8	32	
24		-8X125X200	Q345B	4	1.3	5.2	
23		(-9)x180x308	Q345B	4	3.8	17.2	
22		-8x220x666	Q345B	2	6.7	13.4	
21		-10x160x620	Q345B	2	7.8	15.6	
20		-10x570x730	Q345B	4	29.5	118	
19		-46x245x440	A709-50-2	8	29	232	
18	FCM	-20x816x3380	A709-50-2	4	423	1692	
17		-10x120x520	Q345B	22	4.8	105.6	
16		-12x224x520	Q345B	4	8.7	34.8	
15		-60x635x665	A709-50-2	4	130	520	
14	FCM	-12x380x4004	Q345B	1	110.8	110.8	
13		-10x646x270	Q345B	4	10.5	42	
12		L80X80X8-1940	Q235	2	18.7	37.4	
11		-10x400x3380	Q345B	2	54	108	
10		L80X80X8-1683	Q235	2	16.3	32.6	
9		-10x150x150	Q345B	14	0.92	12.88	
8	FCM Z SteelZ 向板	-30x915x1226	Q345B	4	191	764	
7	FCM Symmetry 对称制作	-20x370x1215	Q345B	4	52	208	
6		-10x526x4160	Q345B	2	145.6	291.2	
5		-10x600x4160	Q345B	2	166.8	333.6	
4	FCM Symmetry 对称制作	-12x280x4000	Q345B	2	105.4	210.8	
3		-12x280x4004	Q345B	1	106.3	106.3	
2		-16x520x850	Q345B	4	55.5	222	
1		-20x300x950	Q345B	4	41.3	165.2	

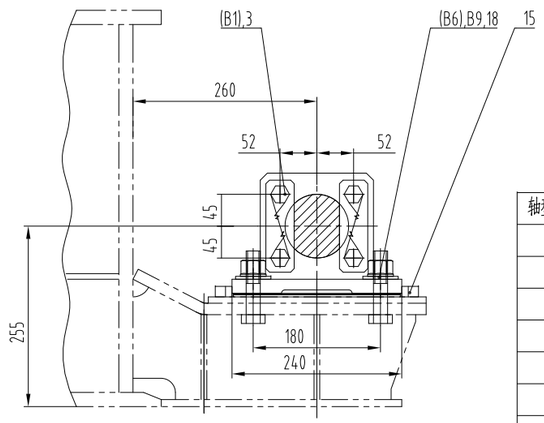
NO.	SUB-NO. or STDD	NAME & SIZE	MTRL or STYLE	Q'TY/SET	SET*WEIGHT	SCALE
	J526A0201(3/3)	Trolley frame 小车架	Welded 焊接件	1	7880.68	1:40

ZPMC	DRAW			TECH.			TRACE		
	CHCK			STDD			CHCK		





B-B



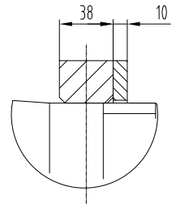
Detail I
1:2



Detail II
1:2



Detail III
1:3



技术要求

1. 安装前,轴承要加适量润滑脂.安装后,再注满润滑脂.
2. 车轮安装调试完毕后,件15焊接牢固.
3. 件16调整垫片组正常使用一片,如需调整,当需轨距减少时,相应增加垫片,反之亦然.
4. 序号17，B10加工，装配，维修时使用,小车正常工作时拆除.
5. 螺栓拧紧好后涂防腐漆.

Technical requirements

1. Fill in appropriate grease oil to bearings before assembling.
   While completing assembly,inject greese full.
2. Weld piece 15 firmly after finishing the Trolley Wheel Assembly.
3. Piece16 use a slice with normal usage .
4. Piece17 and B10 are only used for making,assembling and maintenancing.
   When normally working, depart it .
5. Bolts shall be painted after tightened.

螺栓规格 Bolt spec.	件号 Item No.	等级 Grade	螺栓预紧力(KN) Pretightening force	螺栓预紧力矩(Nm) Pretightening torque
M20	B7,B9	8.8级 8.8s	94.1	270(k=0.145)
M20	B3	8.8级 8.8s	94.1	270(k=0.145)
M16	B1,B5	8.8级 8.8s	60.29	140(k=0.145)

NO.	SUB-NO. OR STDD	VER	NAME & SIZE	MTRL	QTY	EACH	TOTAL	NOTE
B7	GB5782-86		螺栓 M20×130 Bolt	8.8	7	0.4	2.8	'Dacro '
B6	GB889-86		锁紧螺母 M20 Lock nut	8	11	0.0	0.0	'Dacro '
B5	GB32.1-88		螺栓 M16×55 Bolt	8.8	8	0.1	0.8	'Dacro '
B4	SKF(瑞典)		轴承 22230CC/W33 Bearing	外购成品 Pur.	2	18.0	36.0	'Φ150×Φ270×73 'Cr=750KN Cr0=1130KN
B3	GB32.1-88		螺栓 M20×60 Bolt	8.8	4	0.2	0.8	'Dacro '
B2	GB9877.1-88		唇形密封圈 170×200×15 Lip seal	外购成品 Pur.	2	0.0	0.0	
B1	GB32.1-88		螺栓 M16×35 Bolt	8.8	20	0.1	2.0	'Dacro '
19	ZT2-020935-12	0	球铰支承环 spherical bush	外购成品 Pur.	1	9.6	9.6	
18	ZT2-020935-11	0	自制垫圈 Washer	Q235-A	4	0.1	0.4	
17	ZT2-020935-10	0	剖分盖 Cover	Q235-A	1	5.9	5.9	
16	ZT2-020935-09	0	调整垫片组2 Shim Group 2	Q235-A	1	1.6	1.6	
15	ZT2-005111	0	楔形挡块组	Q235-A	2	0.2	0.4	
14	ZT2-020935-08	0	轴套2 Sleeve 2	20	1	1.4	1.4	
13	ZT2-020935-07	0	套 Sleeve	35	1	34.9	34.9	
12	ZT2-020935-06	0	卡轴板 Baffle	Q345-B	2	0.4	0.8	
11	ZT2-020935-05a	0	调整垫片组1 Shim 1	Q235-A	2	0.5	1.0	
10	ZT2-020935-04a	0	支座2 Support2	焊接件 Wel.	1	10.0	10.0	
9	ZT2-020207a	0	轴 Shaft	35CrMo	1	115.3	115.3	
8	ZT2-020935-02	0	压盖2 cover 2	Q345-B	1	6.8	6.8	
7	ZT2-020935-01	0	轴套1 sleeve1	20	1	1.8	1.8	
6	ZT2-020925-04	0	外隔圈 Outer Spacer	Q235-A	1	0.8	0.8	
5	ZT2-020925-03	0	内隔圈 Inner Spacer	Q235-A	1	0.6	0.6	
4	ZT2-020925-02	0	压盖 Cover	Q235-A	1	3.0	3.0	
3	GB4240-84		不锈钢丝 2.0 l=5000 Stainless wire	0Cr18Ni9Ti	1	0.1	0.1	
2	ZT2-020925-01	0	透盖 Through hole Cover	Q235-A	2	7.7	15.4	
1	ZT2-015022	0	小车车轮630x75 Trolley Wheel	42CrMo	1	296.5	296.5	

型号对应表

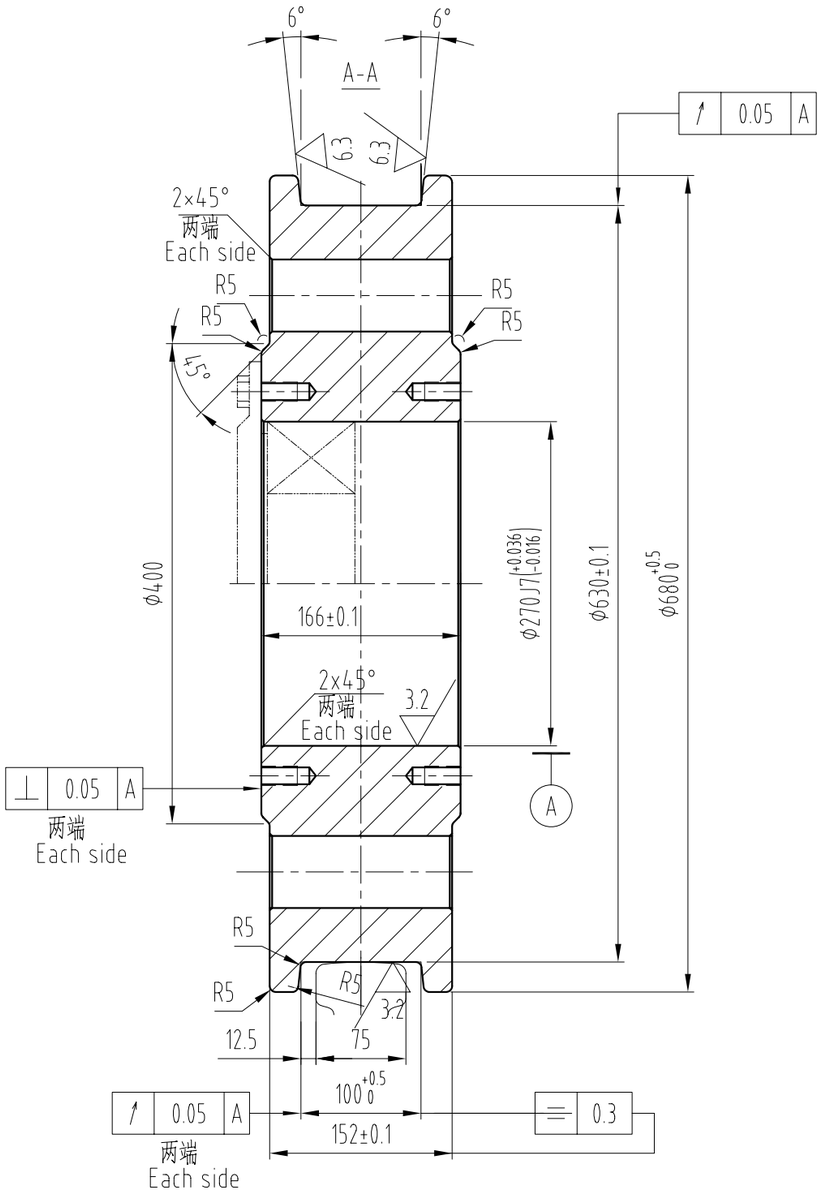
轴型号	油嘴孔尺寸 Oiling screw M
I	Rc1/4''
II	Rc1/8''
III	NPT1/4''
IV	NPT1/8''
V	M10x1
VI	Rp1/4''
VII	Rp1/8''

NO.	SUB-NO. OR STDD	VER	NAME & SIZE	MTRL	QTY	EACH	TOTAL	NOTE
B10	GB5783-86		螺栓 M16×60 Bolt	8.8	2	0.1	0.2	'Dacro '
B9	GB5782-86		螺栓 M20×90 Bolt	8.8	4	0.3	1.2	'Dacro '
B8	GB97.1-85		平垫圈 20 Washer	100HV	7	0.0	0.0	'Dacro '

设计/DSGN	校对/CHKD	审核/EXAM	图幅/FORMAT
张乐	黄小辉		A2
日期/DATE 2011-8-18	日期/DATE	日期/DATE	
材料/MAT 部件 Ass.	比例/SCALE 1:6	图名/DRAWING NAME 小车车轮装配630x75 Trolley Wheel Assembly	
单重/WEI (Kg) 550.1			
上海振华重工集团 ZPMC	图号/DRAWING NO. ZT2-020935-00	版本/REV 0	页号/PAGE 1

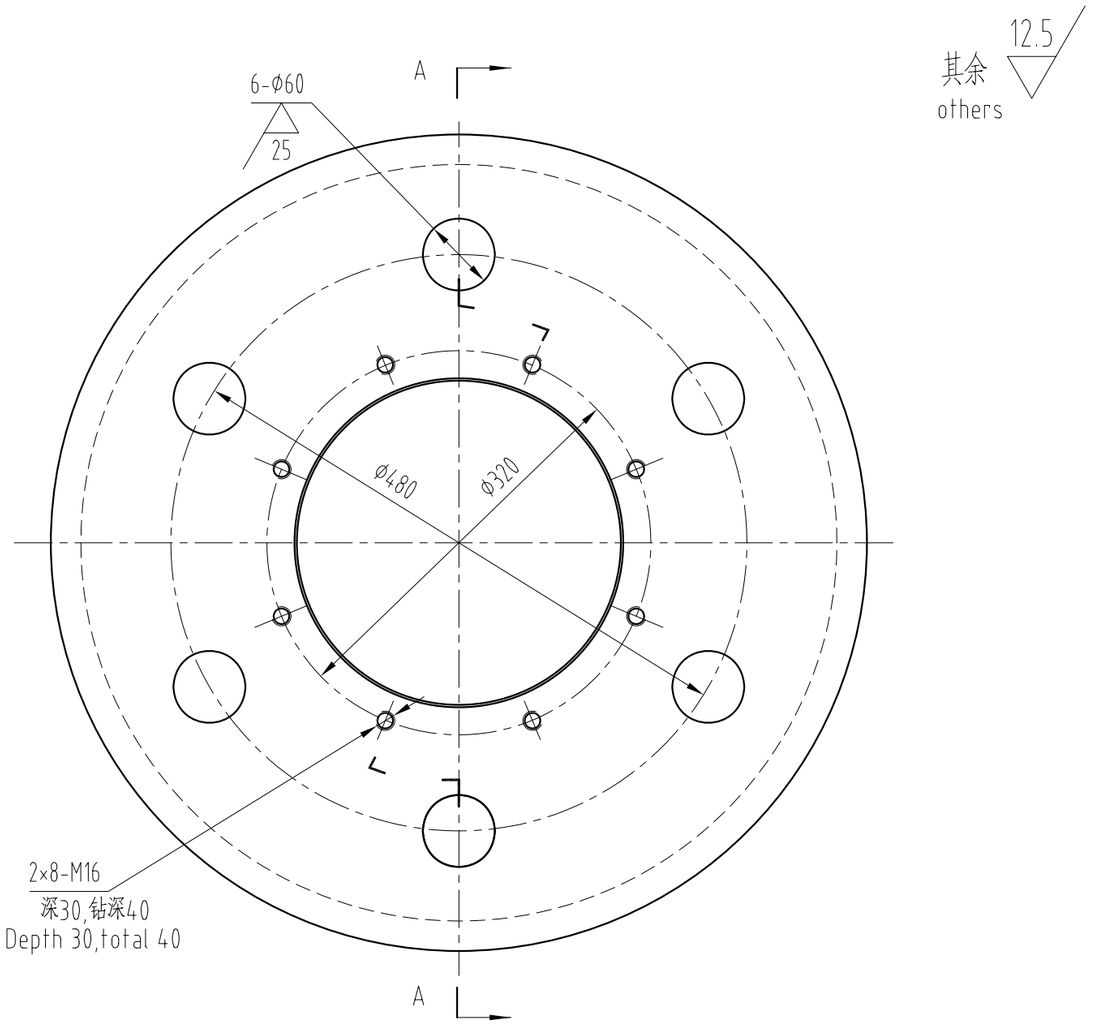
描述及日期/Description & Date　修改标志/Sign

010301t



Technical Requirement

1. Surface heat treatment shall be according to the ZPMC standard which is more critical than the ASTM A504;the hardness of the wheel tread shall be HRC50~56,the effective hardness depth shall not be less than 15mm,hardness at 15mm depth shall not be less than HRC35.
2. The wheel body should be checked and accepted according to the national forging standard GB12361-90 and GB12362-90,and UT inspection should be carried out according to the ASTM A504-C.



技术要求

1. 制作时按公司内控标准(高于ASTM A504标准)制作: 踏面硬度HRC50~56,硬化层有效深度≥15mm,15mm深处硬度≥HRC35.
2. 车轮锻件按GB12361-90和GB12362-90技术条件验收,探伤应按美国ASTM A504-C级进行验收.

描述及日期/Description & Date	设计/DSGN 徐魏魏	校对/CHKD	审核/EXAM	图幅/FORMAT	A3
	日期/DATE 2009-4-28	日期/DATE	日期/DATE		
	材料/MAT 42CrMo	比例/SCALE 1:5	图名/DRAWING NAME 小车车轮630x75 Trolley Wheel		
	单重/WEI (Kg) 296.5				
修改标志/Sign	ZPMC 上海振华重工集团		图号/DRAWING NO. ZT2-015022	版本/REV 0	页号/PAGE 1

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